Brains and Models

1. Structural lens: synapses and weights

A simple model assigns one independent scalar weight to each synapse.

$P \approx S = N \times k$

N = neurons in the chosen region k = mean incoming synapses per neuron

Regional scope. Use N and k from the same region. For a whole-brain estimate, sum N × k across regions and count each synapse once.

Human example. About 86 billion neurons in the brain, about 16 billion in the cerebral cortex. One neocortical study estimated 1.64×10^{14} synapses. 164 Trillion

Limit. This approximation counts connections. Weight sharing, synaptic dynamics and neuron properties change the relationship to model parameters.

Azevedo et al., 2009; Tang et al., 2001. Full citations in speaker notes.

2. Storage lens: bits per synapse and per weight

Evidence. Bartol et al. inferred at least 26 distinguishable strength levels (about 4.7 bits) in sampled rat hippocampal synapses.

This is a lower-bound estimate under that study's assumptions. Applying it to the human brain requires an explicit extrapolation.

Equivalent weight slots ≈ S × b / q

S = synapses b = assumed bits per synapse q = stored bits per weight

What it measures. A nominal storage comparison at a stated precision.

Limits. Stored bit width differs from learned information. Correlations and unequal state probabilities affect capacity. Other biological states add detail.

Bartol et al., 2015. Full citation and assumptions in speaker notes.

3. Graph lens: a description of the wiring

A simple edge list stores both endpoints and a strength for each connection.

Bits per edge ≈ $2\lceil \log_2 N \rceil + b$

With N = 86 billion neurons: 37 bits per endpoint, or 74 address bits.

If b = 4.7, the idealized total is 78.7 bits per edge.
That equals about 4.9 storage words of 16 bits each.

Encoding matters. Adjacency lists, spatial structure and compression can reduce this description length. It is a property of the chosen encoding.

Meaning. These are storage-word equivalents. Dividing by 16 does not turn connection addresses into additional learned parameters.

Address arithmetic is illustrative. Inputs: Azevedo et al., 2009; Bartol et al., 2015.

4. Dynamics lens: reuse of connections over time

Unrolling a recurrent model reuses the same weights at successive time steps.

Connection evaluations ≈ a × S × T

a = fraction of connections evaluated per step T = number of steps

Illustrative example. $S = 1.6 \times 10^{14}$, $T = 100$ and $a = 1$ give 1.6×10^{16} connection evaluations.

Unique weights. The model still has 1.6×10^{14} synaptic weights under the one-weight-per-synapse assumption.

Caveat. T and a depend on the model. A synaptic event has no universal conversion to FLOPs, and biological activity is asynchronous.

Recurrent weight reuse: PyTorch RNN documentation. Biological scale: Tang et al., 2001.

5. Regional lens: a brain area and a model stage

A regional comparison keeps the biological scope closer to the task.

Scope. Choose a region, such as primary visual cortex (V1),
and define which local and long-range inputs count.

Connections. Estimate $S_r = N_r \times k_r$ using compatible regional data.
An ANN with untied scalar weights can match that connection count.

Function. Test predictions of neural responses or behavior.
Matching counts alone does not establish comparable capability.

Biological detail. A neuron can have nonlinear dendrites and changing internal state. Feedback and plasticity affect the comparison.

Azevedo et al., 2009; Beniaguev et al., 2021; PyTorch Conv2d documentation.

Worked example: human neocortical scale

Use $S \approx 1.6 \times 10^{14}$ synapses, rounded from a neocortical study.
For the storage calculation only, assume b = 4.7 bits per synapse.

Comparison	Calculation	Result
Synaptic weights	$P \approx S$	160 trillion weights
16-bit equivalents	$S \times 4.7 / 16$	47 trillion slots
8-bit equivalents	$S \times 4.7 / 8$	94 trillion slots

The first row counts independent weights in an idealized model.
The other rows compare nominal storage under an extrapolated assumption.

Tang et al., 2001 (human neocortex); Bartol et al., 2015 (rat hippocampus).

Comparisons across mammals

1. Define the population. Use the species, brain region and age group relevant to the question. Record sources and uncertainty.

2. Count consistently. Use regional N and incoming k, or measured synapse density × regional volume. Sum regions only with compatible boundaries.

3. Choose the quantity. Report connections, storage bits, wiring description or connection evaluations. Each quantity has different assumptions.

4. Check sensitivity. Vary uncertain inputs. State weight sharing, precision and time horizon before interpreting the result.

For example, about 71 million neurons is a published whole-mouse-brain estimate. A cortical synapse-per-neuron value needs a cortical neuron count.

Herculano-Houzel et al., 2006. Full citation in speaker notes.

Summary: five ways to compare brains and models

Lens	Quantity	What it tells us
Structure	Synapses S	Count of connections
Storage	S × assumed bits per synapse	Nominal storage budget
Wiring	Address bits + strength bits	Description length
Dynamics	Connection evaluations	Work over a defined interval
Region	A brain area and model stage	A functional hypothesis to test

Parameter count, storage and computation answer different questions.
The biological scope and modeling assumptions determine the comparison.

Summary: the human numbers and their limits

≈86 billion neurons. A whole-brain estimate, with about 16 billion in the cerebral cortex. Regional differences matter.

≈160 trillion synapses. A rounded estimate for the human neocortex from one study, rather than a whole-brain total.

≈47 trillion 16-bit slots. A conditional storage equivalent if every synapse contributes 4.7 bits. That assumption extrapolates from rat hippocampus.

These estimates describe different quantities and populations. They do not establish a human-equivalent AI model size.

Azevedo et al., 2009; Tang et al., 2001; Bartol et al., 2015.

Summary: what makes a useful comparison

A clear question. The relevant brain region and task define the scope.

Explicit units. Connections, bits and operations remain separate quantities.

Visible assumptions. Precision, weight sharing and time determine how a biological count maps to a model.

Functional evidence. Neural or behavioral tests are needed to evaluate what a model captures about the brain.

There is no single conversion from synapses to AI capability.